growth forms. A population possesses genetic characteristics that are directly related to their adaptiveness, reproductive success, and persistence in their habitats over time. Life history of an organism is an important part of this attribute. The population has a definite structure and function that can be described with reference to time.

10.8 Population Attributes

Population density

The density of a population refers to its size in relation to unit of space and time. Population density is the total number of that species within a natural habitat. The size of the population can be measured in several ways, including abundance (absolute number in population), numerical density (number of individuals per unit area (or) volume) and biomass density (biomass per unit area (or) volume). The population density of a species can also be expressed with reference to the actual area of habitat available to the species When the size of individuals in the population is relatively uniform then density is expressed in terms of number of individuals (numerical density).

Natality

Populations increase because of natality. Natality is equivalent to birth rate and is an expression of the production of new individuals in the population by birth, hatching, germination (or) fission. The two main aspects of reproduction, namely fertility and fecundity play a significant role in a population. Natality rate may be expressed in crude birth rate number of organisms born per female per unit time.

$$\text{Birth rate (b)} = \frac{\text{number of birth per unit time}}{\text{average population}}$$

Mortality

Mortality is the population decline factor and is oppposite to natality. Mortality can be expressed as a loss of individuals in unit time or death rate. Generally, mortality is expressed as specific mortality, that is, the number of members of an original population dying after the lapse of a given time. The crude death rate of a population can be calculated by the equation.

$$\text{Death rate (d)} = \frac{\text{number of deaths per unit time}}{\text{average population}}$$

The rate of mortality (death) is determined by density. Mortality is high at high density because of the hazards of overcrowding, increased predation and spread of disease.

Mortality rates vary among species and are correlated and influenced by a number of factors such as destruction of nests, eggs or young by storms, wind, floods, predators, accidents and desertion by parents.

Population Dispersion

Populations have a tendency to disperse or spread out in all directions, until some barriers are reached. This is observed by the migration of individuals into (Immigration) or out (Emigration) of the population area.

Migration

Migration is a peculiar and unique kind of mass population movement from one place to another and back. To avoid the severe winter cold, Siberian cranes migrate from Siberia to Vedanthangal in Tamil Nadu and return back in spring. Some fishes are known to migrate from sea to fresh water (anadromous migration, Salmon) and some from fresh water to sea (catadromous migration, Eel).

Emigration

Under natural conditions, emigration usually occurs when there is overcrowding. This is regarded as an adaptive behavior that regulates the population in a particular site and prevents

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